NOT MEASUREMENT SENSITIVE

MIL-DTL-24641B
4 May 2015
SUPERSEDING
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27 October 1986

MILITARY SPECIFICATION

ACID, HYDROFLUORIC, TECHNICAL

Inactive for new design after 8 April 2010.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a technical grade of hydrofluoric acid.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation – VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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FEDERAL STANDARD

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations (CFR), Title 49
49 CFR 171.1 – Department of Transportation Rules and Regulations for Transportation of Explosives and Other Dangerous Articles.

DEPARTMENT OF LABOR
Occupational Safety and Health Administration (OSHA)
Code of Federal Regulations
29 CFR, Part 1910.1200 – Hazard Communication Standard.

(Copies of these documents are available online at http://www.gpoaccess.gov/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI Z1.4 - Sampling Procedures and Tables by Inspection of Attributes

(Copies of ANSI/ASQ Z1.4 are available online at www.asq.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Hydrofluoric acid, technical, shall contain not less than 60 percent of hydrogen fluoride (HF) by weight, in water solution, when tested as specified in 4.3.1.

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3.2 Labeling. Each container shall be durably and legibly labeled with the following:

Hydrofluoric Acid

Danger

Hazardous liquid and vapor causes severe burns which may not be immediately painful or visible.

VAPOR will burn eyes, nose and throat, HARMFUL IF INHALED. LIQUID will burn skin and eyes, HARMFUL IF SWALLOWED. Do not get in eyes, on skin or clothing. Do not breathe vapor. Keep container closed when not in use. Use with adequate ventilation. Wash thoroughly after handling. Store out of sun and away from direct heat.

FIRST AID

VAPOR EXPOSURE: Move to fresh air. If breathing has stopped, give artificial respiration. If breathing is difficult, give oxygen. If in eyes, flush with plenty of water for at least 30 minutes. Get medical attention.

LIQUID EXPOSURE: Always have on hand a supply magnesium sulfate solution (minimum 25% concentration). Remove all contaminated clothing. Flush affected areas with plenty of water. (Avoid high pressure water streams to prevent further damage to already burned skin.) Place the areas in a cool solution of 25 % magnesium sulfate. If the affected body area is large or on the trunk, cool compresses saturated with the 25 % magnesium sulfate solution should be used. If in the eyes, flush with plenty of water for at least 30 minutes. If swallowed and victim is conscious, have victim drink water or milk. DO NOT INDUCE VOMITING. Seek medical attention for any exposure to HF. Wash clothing before reuse.

3.2.1 Safety data sheet. The contracting activity shall be provided a safety data sheet (SDS) at the time of the contract award. The SDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify, and supplement FED-STD-313.

4. VERIFICATION

4.1 Conformance inspection.

4.1.1 Sampling procedure for conformance tests. From each lot, three separate 1-quart samples shall be taken. The samples shall be taken to represent respectively, the first, middle and last part of the run which produced the lot. Samples shall be taken only in containers made of steel, lead, hard rubber, cerasin, or polyethylene, since the acid covered by this specification is highly corrosive and attacks other metals and glass. Each sample shall be thoroughly mixed and

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divided into three equal portions. The portions shall be placed in clean, dry containers which shall be sealed and carefully marked. The three portions of the sample shall be used as follows:

a. One for quality conformance inspection tests.
b. One for the contractor.
c. One to be held by the Government to be used for retests in case of dispute.

4.1.2 Sampling for inspection of containers. A random sample of packed containers from each lot shall be taken in accordance with ANSI Z1.4 at inspection level I. Acceptance criteria shall be as specified in the procurement document (see 6.2).

4.2 Retests and rejections. Failure of the material to conform to the requirements of this specification during inspection shall be cause for rejection of the lot represented. If, in the opinion of the contractor, the failure was due to faulty test methods, the tests may be repeated on the contractor's retained sample. In case of a controversy, the test shall be repeated on the Government retained sample. Failure of any two of the three samples to meet the requirements shall be cause for final rejection of the lot represented. Material which has been rejected may be reworked to correct the defects and resubmitted for acceptance. Information concerning previous rejection and the action taken to correct the defects found in the original material shall be furnished before resubmitting material.

4.3 Test procedures.

4.3.1 Hydrogen Fluoride.

4.3.1.1 Apparatus. The following apparatus shall be used in the testing procedures.

a. A 100-milliliter (mL) burette.
b. Platinum weighing tube (10 mL capacity), fitted with a platinum cap.
c. Two platinum dishes (125 mL capacity).
d. Hot plate.
e. Steam bath.
f. Gravimetric balance.

4.3.1.2 Reagents. Reagents shall consist of standardized 1 normal sodium hydroxide (NaOH) solution, which shall be kept in a cerasin-lined container to prevent contamination with silica; standardized 1/10 normal iodine solution; a saturated solution of potassium nitrate (KNO_3); phenolphthalein indicator; and starch indicator.

4.3.1.3 Procedure for determining total acidity of hydrofluorosilicic acid. A portion of 3 grams of the sample shall be accurately weighed in a platinum weighing tube. Ten mL of a saturated solution of KNO_3 shall be poured into a platinum dish of approximately 125 mL capacity and chopped ice added. Fifty mL of standard 1 normal NaOH shall be run in from a burette and 3 drops of phenolphthalein indicator added. The weighing tube containing the weighed sample shall be immersed beneath the surface of the caustic and the cover slowly removed with a platinum wire to allow the sample to mix slowly with the caustic. The solution

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shall be kept cold and titrated with the standard NaOH in the burette to the first permanent pink color. The total milliliters used shall be noted as (A). The dish shall be put on a hot plate and the solution warmed to 176 degrees Fahrenheit (°F) (80 degrees Celsius (°C)). Additional standard NaOH shall be added to obtain the second end point and the additional milliliters noted as (B).

4.3.1.4 Procedures for determining sulfuric acid in hydrofluoric acid. A portion of 10 grams of the sample shall be accurately weighed in a platinum weighing tube and transferred to a platinum dish. The weighing tube shall be rinsed into the dish with distilled water. The solution shall be evaporated to constant volume (near dryness) on a steam bath. A few drops of water shall be added and the evaporation repeated. Lack of odor during the second evaporation indicates that all acids other than sulfuric have been expelled. The residue shall be cooled, dissolved in 100 mL of carbon dioxide (CO_2) free water, and titrated with standard 1 normal NaOH solution, using phenolphthalein indicator. The milliliters needed shall be noted as (C).

4.3.1.5 Procedure for determining sulfurous acid in hydrofluoric acid. A portion of 10 grams of the sample shall be accurately weighed in a platinum dish with 75 mL of water. This solution shall be titrated with standard 1/10 normal iodine using starch indicator. The milliliters used shall be noted as (D).

4.3.1.6 Calculation of the percentage of hydrogen fluoride. The percentage of hydrogen fluoride may be calculated by the following formula:

$$\text{Percentage of HF} = N_S\left[\frac{2A\text{-}B}{S_1} - \frac{2C}{S_2}\right] - \frac{2DN_1}{S_3}$$

Where A, B, C, and D are the milliliters used and S_1, S_2, and S_3 are the samples weight taken in the procedures specified in 4.3.1.3, 4.3.1.4, and 4.3.1.5, respectively. N_S and N_1 are the normalities of the sodium hydroxide and iodine solutions, respectively.

4.3.2 Labeling inspection. The packaging shall be inspected for labeling as specified in 3.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hydrofluoric acid, technical grade is intended for use in removing sand particles from metallic castings and for etching glass.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Acceptance criteria (see 4.1.2).
c. Packaging requirements (see 5.1).
d. Hazardous material identification and SDS (see 6.5).
e. Activities requiring a copy of the SDS (see 6.5).

6.3 Transportation description. Transportation description applicable to these items are:

Acid, hydrofluoric.
Carload weight 36,000 pounds.
Truckload weight 36,000 pounds.

6.4 Purchasing unit. Hydrofluoric acid should be purchased by volume, the unit being a U.S. gallon, 231 cubic inches at 77°F (25°C).

6.5 Safety data sheet (SDS). Contracting officers will identify those activities requiring copies of the completed SDS prepared in accordance with FED-STD-313 and meeting the requirements of 29 CFR 1910.1200. ~~The pertinent government mailing addresses for submission of the data are listed in Appemdix B of FED-STD-313.~~ Contracting officers will identify the activities requiring copies of the SDS (see 6.2).

6.6 Definitions.

6.6.1 Lot. A sampling lot will consist of not more than 16,000 pounds of hydrofluoric acid offered for delivery at one time. Material in a lot will be identified by order of production until ultimate action is taken as to the acceptance or rejection of the lot.

6.7 Subject term (key word) listing.

Hydroflourosilicic acid
Hydrogen fluoride
Sulfuric acid
Sulfurous acid

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
DLA - GS

Preparing Activity:
DLA - GS3

(Project 6810-2015-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

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